

Research report

Gastrodin ameliorates depression-like behaviors and up-regulates proliferation of hippocampal-derived neural stem cells in rats: Involvement of its anti-inflammatory action



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HIGHLIGHTS

- We investigated early intervention effects of gastrodin on CUS rats' behaviour.
- It reduced depressive-like behaviour and promoted hippocampal and stem cell proliferation.
- It also inhibited NF- κ B protein expression and IL-1 β level in the hippocampus.
- Moreover, it protected the viability of NSCs from IL-1 β -induced damage.
- The findings support gastrodin as a newly antidepressant.

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ABSTRACT

Gastrodin (GAS), an active constituent of the Chinese herbal medicine tianma, has antidepressant-like activity in animals but no specific molecular mechanisms have been identified. In the present study, chronic unpredictable stress (CUS) was used to establish a rat depression model; The sucrose preference test, forced swim test and Morris water maze test were used to assess depression-like behaviors (anhedonia, behavioral despair, motor retardation, and poor spatial memory), and the proliferation of hippocampal stem cells was tested by BrdU immunohistochemistry. The stress and inflammatory responses were assayed by measuring IL-1 α , NF- κ B, and p- κ B expression by Western blot and IL-1 β production by ELISA. Direct and indirect effects of GAS on NSC viable cell number were examined *in vitro* by WST-1 and BrdU assays. It was found that GAS (200 mg/kg daily) reversed all tested depression-like behaviors in CUS model rats and up-regulated NSCs proliferation in the hippocampus. Enhanced expression of p- κ B, NF- κ B, and IL-1 β by CUS was also reversed by GAS. Moreover, *in vitro* experiments revealed that GAS alone did not increase the viability of NSCs but protected them from IL-1 β -induced damage. These results support the antidepressant and neuroprotective effects of GAS, and GAS may reduce depression-like behaviors by protecting hippocampal NSCs against the proinflammatory cytokine IL-1 β .

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1. Background

According to the World Health Organization, depression will rank as the second most common cause of disability by 2020. Antidepressant drugs are the most common first-line treatments for depression. Current pharmacological treatments, including monoamine oxidase inhibitors (MAOI), tricyclic antidepressants (TCA), selective serotonin reuptake inhibitors (SSRIs), serotonin-norepinephrine reuptake inhibitors (SNRIs), and noradrenaline-serotonin specific antidepressants (NaSSAs) are generally effective,

Abbreviations: GAS, Gastrodin; CUS, chronic unpredictable stress; NSCs, neural stem cells; SPT, sucrose preference test; FST, forced swim test; MWM, Morris water maze; BrdU, Bromodeoxyuridine; NF- κ B, nuclear factor- κ B; IL-1R1, IL-1 receptor type 1; IL-1RA, IL-1 receptor antagonist; p- κ B, phospho-inhibitor of kappa B.

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but up to one-third of patients may still show limited response, relapse [1], or drug intolerance [2]. Depression comorbid with another psychiatric disorder such as anxiety may also limit treatment response [3]. Addition of atypical antipsychotics is one treatment option to enhance response rates, but such polypharmacy is also associated with decreased compliance due to side effects [4]. Treatments such as deep brain stimulation (DBM), ECT, or repetitive transcranial magnetic stimulation (rTMS) may be effective for some of these drug-refractory patients, but invasiveness (DBM), low tolerability (ECT), or modest efficacy (rTMS) limit broad application of these techniques [5]. Thus, improved pharmacotherapies are still required to substantially reduce the global burden of major depression.

The development of traditional antidepressants was based on the theory that depression arises from a deficit in brain monoamines or monoaminergic transmission [6]. There is a growing consensus, however, that this theory is insufficient to explain the therapeutic effects of antidepressants [7–9]. Indeed, while these drugs alter synaptic levels of monoamines almost immediately, clinically significant effects are generally not observed for several weeks to months, suggesting that these agents trigger gradual changes in neurochemistry, synaptic transmission, postsynaptic signaling, protein expression, and (or) structural neuroplasticity. The inflammatory and neurodegenerative hypothesis of depression has been formulated gradually [10,11] in parallel with preclinical studies from both animals and humans demonstrating neurodegeneration and reduced neurogenesis in depression caused, at least in part, by inflammation and cell-mediated immune activation. It has even been proposed that depression evolved as part of a complex system of immune defenses against environmental pathogens. Animal models have shown that depression-like behaviors are associated with cell-mediated immune responses [12]. Clinically depressed patients manifest higher levels of inflammatory biomarkers, while proinflammatory cytokines induce neuropsychiatric symptoms as well as major depressive episodes [13,14]. Animal studies have found that IL-1 β expression by hippocampal neural progenitor cells and the nuclear factor- κ B (NF- κ B) signaling pathway are critical for the anti-neurogenic and depression-like behaviors caused by acute and chronic stress [15]. These observations identify neurogenesis and neuroinflammation as potential targets for a newer generation of clinical antidepressants.

Gastrodia elata Blume, known as tianma in Chinese, is a plant of the orchidaceae family, the tuber of which can be dried and used medicinally. According to theories of traditional Chinese medicine (TCM), tianma acts to pacify the liver and “calm wind”. Pharmacological studies indicate that tianma has anti-convulsive, sedative, analgesic, nootropic, and anti-inflammatory effects, and improves microcirculation and general circulatory functions; in addition, it is often used in TCM for the treatment of cardio-, cerebro-, and neurovascular diseases including migraine, epilepsy, and depression [16–19]. The phenolic glucoside gastrodin (GAS) is a main active constituent of tianma. It is reported that GAS could inhibit expression of inducible NO synthase, cyclooxygenase-2 and pro-inflammatory cytokines in cultured LPS stimulated microglia [20]; In addition, our previously studies showed that GAS could ameliorates the depressive-like behaviors of CUS rats [21], and it could reverse the levels of IL-6 and IL-1 β , and the expression of iNOS and the p38 MAPK phosphorylation in the hippocampus of enhanced single prolonged stress-induced posttraumatic stress disorder rats model [22], suggesting that GAS may suppress specific signaling pathways associated with the immune response, thereby maintaining hippocampal function and preventing behavioral signs of depression. To test this possibility, we investigated the potential neuroprotective efficacy of GAS and its effects on stress related factors NF- κ B and IL-1 β in the chronic unpredictable stress model of depression. We also observed the effects of GAS on

NSC proliferation in the hippocampus, as reduced NSC proliferation is known to occur in depression while antidepressant treatments reverse this effect [23–25].

2. Materials and methods

2.1. Animals

Eight-week-old male Sprague–Dawley rats were obtained from the Fourth Military Medical University animal center and maintained under standard laboratory conditions (12-h light/12-h dark cycle with lights on at 8:00 AM, $22 \pm 2^\circ\text{C}$ with a relative humidity at $50 \pm 10\%$, food and water ad libitum). The animals were allowed to adapt to laboratory conditions for at least one week. All stress-exposed rats were singly housed while non-stressed rats were grouped four/cage. All procedures were in strict accordance with the guidelines established by the US National Institutes of Health (NIH) and approved by the Fourth Military Medical University Animal Care Committee.

2.2. Experimental procedures

After the adaptive phase, rats were randomly divided into 8 groups of 12: sham, sham + GAS (L) (low-dose, $L = 50\text{ mg/kg/day}$), sham + GAS (M) (medium dose, $M = 100\text{ mg/kg/day}$), sham + GAS (H) (high-dose, $H = 200\text{ mg/kg/day}$), CUS, CUS + GAS (L), CUS + GAS (M), and CUS + GAS (H). From week 1 to week 5, the 4 CUS groups were exposed to two stressors per day in random sequence to maximize the unpredictability. The CUS procedures were performed as described [26]. At week 6, rats were treated with saline or GAS once daily for 14 days. Following drug treatments, individual treatment groups were subdivided for behavioral testing and Western blot and Elisa analysis or BrdU immunohistochemistry: All the rats in each group were exposed to sucrose preference test, and 8 of them were then exposed to forced swim test and sacrificed for BrdU detection or Western blotting and Elisa subsequently; the remaining ($n = 4$) were exposed to Morris water maze.

To examine the direct effects on NSCs, different doses of GAS were administered to cultured NSCs for 3 or 5 days and viable cell number estimated by WST-1 assay. To investigate the protective effect of GAS against IL-1 β , this cytokine (10 ng/ml) was added to the cell medium for 2 h followed by GAS treatment for 24 or 48 h. Cell number was estimated by WST-1 and BrdU assays.

2.3. Administration of GAS

GAS (purity > 99.2%) was supplied by Kunming Pharmaceutical Corporation (Kunming, China). Solubility was >300 mg/mL and bioactivity was stable for more than 2 years at room temperature in sterile water. In the present study, GAS was dissolved in saline and administered intraperitoneally to all GAS groups at the same time. The doses were chosen according to other pharmacological studies in rats [27,28] and adjusted according to our preliminary tests. In the *in vitro* study, GAS was dissolved in DMEM/F12 medium, and then diluted in DMEM/F12 to yield $10\times$ stocks for final treatment concentrations of 5, 10, 20, and 50 $\mu\text{g/mL}$. The vehicle in the sham group was DMEM/F12 medium.

3. Behavioral testing

3.1. Sucrose preference test

The sucrose preference test (SPT) was used as indicator of anhedonia (lack of interest in rewarding stimuli), which is present in some forms of affective disorder, including depression. Specially,

anhedonia was defined as a reduction in sucrose preference over water relative to the control group. Testing was carried out in the home cage using our previously described procedure [29]. Briefly, rats were housed and habituated to a 1% (wt/vol) sucrose solution for 48 h to avoid neophobia during testing. On test days, after 4-h water deprivation, rats were presented with 2 dual bearing sipper tubes containing 1% sucrose or tap water for 1 h, and intake volumes were measured in ml. To prevent location preference, the bottle positions were changed daily. Sucrose preference was calculated as (%) = sucrose consumption/(sucrose consumption + water consumption).

3.2. Forced swim test

The forced swim test (FST) is one of the most common animal tests for depression-like behavior [30]. The FST assesses the degree of learned helplessness or behavioral despair by measuring passive floating in an inescapable fluid-filled cylinder. Twenty-four hours after a pretest session (10 min), the FST was conducted by placing rats in cylinders filled with water (23 °C to a depth of 30 cm) for 5 min and scored as described [31]. The animals were considered floating when three paws were immobile and the fourth paw exhibited only minimal movement. Learned helplessness behavior was defined as an increase in floating time.

3.3. Morris water maze

Spatial learning and memory performance was measured in the Morris water maze (MWM) according to the classic Morris protocol [32]. Spatial learning was evaluated from video records acquired during training trials and the probe test. The rat was placed tail-first in a quadrant not containing the submerged escape platform with head facing the wall. The animal was allowed to swim for 60 s. If it failed to locate the escape platform within this time, it was gently assisted onto the platform and allowed to remain there for an additional 20 s. Animals received 4 learning trials each day from a semi-random starting quadrant (i.e., one not containing the platform) for 5 consecutive days. Then spatial memory was measured on day 6 in a probe trial as the ratio of time spent in the former platform (target) quadrant and the locomotor activity was calculated as the average swimming speed of training trials in the first day by using an automatic tracking system (Topscan, Clever Sys Inc., USA).

3.4. NSC culture and identification

The effect of GAS on neural stem cell (NSC) proliferation was assessed *in vitro* as described in our previous study [33]. Briefly, hippocampal tissue was isolated from brains of embryonic day 14.5 (E14.5) and E16.5 rats under a stereomicroscope. Single cell suspensions obtained by mechanical dissociation were cultured in serum-free Dulbecco's modified Eagle's medium (DMEM) supplemented with nutrient mixture F-12 Ham (F12) containing 20 ng/ml bFGF (100–18B, PeproTech, USA), 20 ng/ml epidermal growth factor (400–25, PeproTech, USA), B-27 and N-2 supplements (Gibco, USA), penicillin, and streptomycin. After one week *in vitro*, primary neurospheres were passaged.

To confirm the NSC phenotype by nestin immunostaining, neurospheres were first collected and post-fixed in 4% paraformaldehyde for 2 h at 4 °C and then incubated in 0.1 M phosphate buffer with 20% sucrose at 4 °C for cryoprotection overnight. Neurosphere sections (10 µm) were cut frozen on a cryostat and mounted onto gelatinized slides for immunostaining. Alternatively, single NSCs were plated onto the poly-L-lysine coated coverslips. After 6 h of attachment, cells were fixed in 4% paraformaldehyde in phosphate buffered saline (PBS). Slides and coverslips were rinsed

in PBS, incubated in mouse anti-nestin antibody for 12 h (ab11306, 1:500, Abcam, Hong Kong) followed by anti-mouse Alexa Fluor 594-conjugated secondary antibody for 2 h (A21203, 1:500, Invitrogen), and then counterstained with 4,6-diamidino-2-phenylindole (DAPI) for 20 min at room temperature. Stained specimens were mounted in PBS/glycerol (vol/vol) and observed under a fluorescence microscope.

3.5. BrdU detection

To label newly proliferated cells *in vivo*, rats (4 per group) were given an intraperitoneal injection of bromodeoxyuridine (BrdU, 100 mg/kg) once a day for 3 days from the last day of saline or GAS treatment. After sacrifice perfusion with 100 mL of 0.1 M cold PBS and 400 mL 4% cold PBS-buffered paraformaldehyde, brain samples were soaked in 25% sucrose paraformaldehyde at 4 °C for 1 h. Fixed brain samples were cut at 20 µm and incubated in 2N hydrochloric acid for 1 h and in 0.1 M sodium borate (pH 8.5) for 10 min. Mouse anti-BrdU antibody (B8434, 1:500, Sigma–Aldrich) and anti-mouse Alexa Fluor 594-conjugated secondary antibody were applied in sequence and sections observed under a fluorescence microscope.

To label the NSCs *in vitro*, 10 µM BrdU were given for 24 or 48 h, then they were dissociated into single cells and plated onto the poly-L-lysine coated coverslips at a concentration of 3×10^4 per well in 24-well plates. Cultures were immunostained by sequential overnight incubation with mouse anti-BrdU, incubation with Alexa Fluor 488-conjugated secondary antibody (A21202, 1:500, Invitrogen), and counterstaining with 50 µg/mL propidium iodide (PI; P4170 Sigma–Aldrich, USA). The BrdU-labeled cells were counted using a fluorescence microscope and normalized to the total cell number of PI-stained cells.

3.6. Determination of cell viability

Cell viability was determined by the 2-(4-Iodophenyl)-3-(4-nitrophenyl)-5-(2,4-disulphophenyl)-2H-tetrazolium monosodium salt (WST-1, Roche) assay. Neural stem cells were seeded at 1×10^4 cells/well in 96-well plates. After treatment with GAS, WST-1 was added to each well for 4 h at 37 °C. The optical density was measured at 450 nm by a microplate reader. Data are presented as averages obtained from three independent experiments with at least 5 replicates (mean $\pm$ S.D., $n = 5$).

3.7. Western blotting

Tissues and cells were lysed with SDS-PAGE sample buffer composed of 62.5 mM Tris–HCl, 2% w/v SDS, 10% glycerol, 50 mM DTT, and 0.1% w/v bromophenol blue, and the insoluble materials separated by centrifugation at $12,000 \times g$ for 10 min. The supernatant was heated to 100 °C for 10 min then and cooled on ice for 30 min. Protein extracts were separated by SDS polyacrylamide gel electrophoresis and then transferred onto PVDF membranes (Millipore). Membrane was blocked with 5% skim milk in Tris buffered saline (TBS) and then incubated overnight at 4 °C with rabbit anti-IL-1 receptor antagonist (IL-1RA, 1:1000), anti-IL-1 receptor type 1 (IL-1R1, 1:1000), rabbit anti-phospho-inhibitor of kappa B (pIkB, 1:2000), rabbit anti-NFκB (1:2500) or mouse anti-β-actin. After washing, membranes were incubated with a horseradish peroxidase (HRP)-conjugated goat anti-rabbit or goat anti-mouse IgG for 1 h at room temperature. The antibody-reactive bands were visualized on X-ray film using super ECL plus detection reagent (Supersignal west pico chemiluminescent substrate, Thermo, USA, 34077).

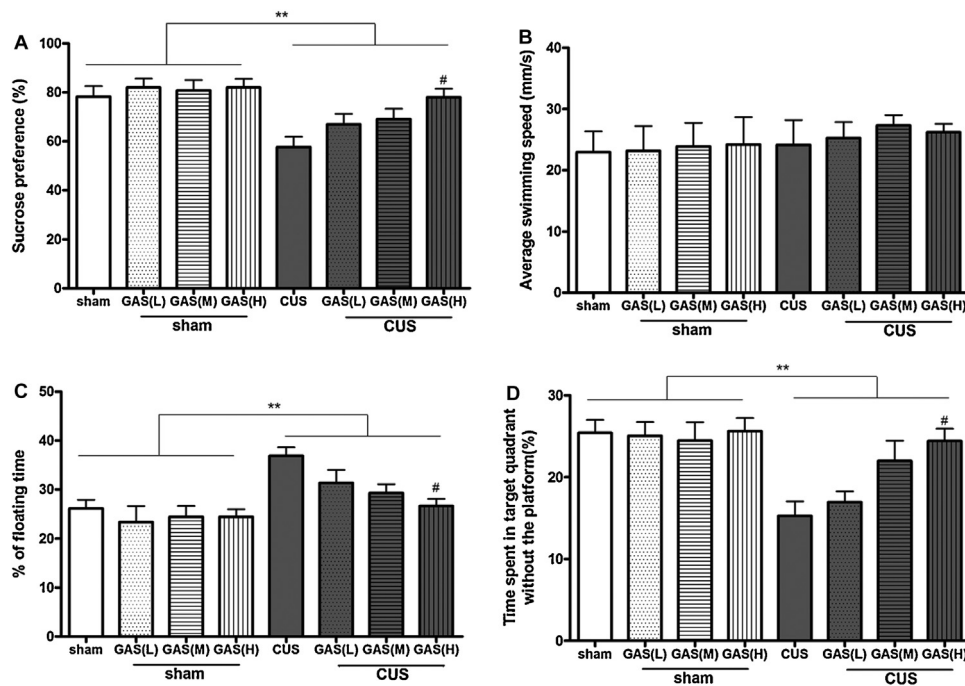


Fig. 1. Reversal of CUS-induced depression-like behaviors (anhedonia, learned helplessness, and motor retardation) and impaired spatial memory by gastrodin (GAS). (A) Mean sucrose consumption in the sucrose preference test, a measure of anhedonia. GAS dose-dependently increased sucrose consumption. (B) There is no significant difference in the average swimming speed of MWM training trials in the first day. (C) Floating time in the forced swim test revealed that CUS-induced learned helplessness was reversed dose-dependently by GAS. (D) Time spent in the target quadrant during the MWM probe trial indicated that the CUS-induced spatial memory deficit was reversed dose-dependently by GAS. Data were analyzed by two-way ANOVA with repeated measurement, followed by a Student–Newman–Keuls *post hoc* test. ** $P < 0.01$, # $P < 0.05$ vs. CUS group.

3.8. IL-1 β measurement by Elisa assay

Supernatants of hippocampal tissue homogenates from each treatment group were collected and the levels of IL-1 β detected by a Rat Elisa Kit according to the manufacturer's instructions (SunRed, 0120, China).

3.9. Statistics

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS Inc., Chicago, IL, USA). Data were subjected to one-way or two-way analysis of variance (ANOVA) followed by Student–Newman–Keuls or Least Significant Difference *post hoc* tests for between-group comparisons. Differences were considered significant when $P < 0.05$.

4. Results

4.1. GAS treatment reversed chronic unpredictable stress (CUS)-induced depression-like behaviors and impaired spatial memory

As shown in Fig. 1, two-way ANOVA revealed significant differences of stress factor (sham vs. CUS) in sucrose consumption ($F_{1,88} = 20.807$, $P < 0.01$), floating time in the FST ($F_{1,56} = 19.21$, $P < 0.01$) and the percentage time spent in the target quadrant in the MWM ($F_{1,24} = 18.61$, $P < 0.01$). There were also significant differences of GAS dose factor (vehicle vs. RA (L) vs. RA (H) vs. RA (H)) in sucrose consumption ($F_{3,88} = 3.092$, $P = 0.036$), floating time ($F_{3,56} = 2.896$, $P = 0.043$), and time spent in the target quadrant ($F_{3,24} = 2.829$, $P = 0.047$), and there were no significant differences of stress factor ($F_{1,24} = 0.461$, $P = 0.453$) and GAS dose factor ($F_{3,24} = 0.245$, $P = 0.864$) in the average swimming speed in the MWM, indicating a dose-dependent effect of GAS on depression-like behaviors. In addition, there were no significant differences of

interaction effect (stress $\times$ GAS dose, $P > 0.05$) in these parameters. *Post hoc* comparisons further showed that there were significant differences in both depression-like behaviors and spatial memory performance between rats receiving high dose GAS (200 mg/kg daily) and rats receiving vehicle ($P < 0.05$ for all). Thus, GAS dose-dependently reversed both CUS-induced depression-like behaviors and the spatial memory deficit.

4.2. GAS reversed CUS-induced expression of inflammatory cytokines

Expression levels of IL-1 α , NF- κ B, p-I κ B, and IL-1R1 were examined by Western blot and IL-1 β production by ELISA (Fig. 2). Two-way ANOVA revealed that there was no significant difference of stress factor (sham vs. CUS) in the expression of IL-1R1 ($F_{1,32} = 0.021$, $P = 0.886$), but there were significant differences of stress factor in IL-1 α ($F_{1,32} = 18.37$, $P < 0.01$), NF- κ B ($F_{1,32} = 9.539$, $P < 0.01$), p-I κ B ($F_{1,44} = 4.928$, $P = 0.032$), and IL-1 β ($F_{1,56} = 58.163$, $P < 0.01$). Furthermore, there were also significant differences of GAS dose factor in NF- κ B ($F_{3,32} = 3.481$, $P = 0.027$) and IL-1 β ($F_{3,56} = 3.175$, $P = 0.031$). *Post hoc* comparisons showed significant differences in NF- κ B between rats receiving medium and high-dose GAS (100 or 200 mg/kg daily) compared to rats receiving vehicle ($P < 0.05$), as well as in hippocampal IL-1 β production between the high-dose GAS group and vehicle-treated rats ($P < 0.05$). Thus, in addition to reversal of behavioral signs of depression and cognitive disruption, GAS suppressed the hippocampal expression of inflammatory mediators.

4.3. GAS reversed CUS-induced suppression of hippocampal NSC proliferation

Incorporation of BrdU (BrdU $^{+}$ cells) in the dentate gyrus (DG) of the hippocampus was used to assess the effects of CUS and subsequent GAS treatment on proliferation of presumed NSCs. As shown

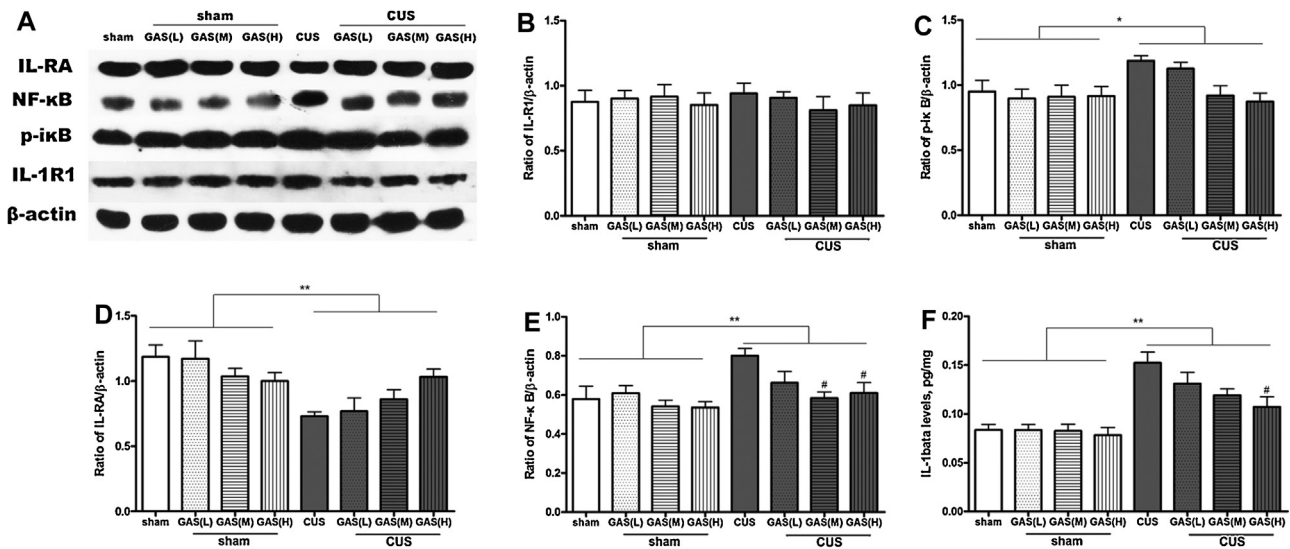


Fig. 2. Reversal of CUS-induced enhancement of inflammatory mediators by GAS. Evaluation of IL-1R1, p-IκB, IL-1RA, and NF-κB protein levels in each treatment group by Western blot ($n=5$ each group) and IL-1β production by ELISA (A) Representative bands for each group. (B–E) Densitometric analysis for IL-1R1, p-IκB, IL-1RA, and NF-κB, respectively. (F) IL-1β for each group as measured by ELISA. Data were analyzed by two-way ANOVA with repeated measurement, followed by a Student–Newman–Keuls *post hoc* test. ** $P<0.05$, # $P<0.05$ vs. CUS group.

in Fig. 3, two-way ANOVA revealed that there were significant differences of both stress factor ($F_{1,56} = 30.477$, $P<0.01$) and GAS dose factor ($F_{3,56} = 2.842$, $P=0.049$) in the number of BrdU⁺ cells in the DG. *Post hoc* comparisons showed significant differences between the CUS + H-GAS and CUS + vehicle group ($P<0.05$). There were no significant differences in the number of BrdU⁺ cells in the DG among the sham, sham + GAS (L), sham + GAS (M), sham + GAS (H) groups.

4.4. Effect of GAS and IL-1β on NSC proliferation in vitro

Under isolation and culture conditions favoring NSC survival from fetal (E14.5) hippocampal tissue, the majority of re-plated cells expressed the NSC marker nestin (Fig. 4A). There were no significant differences in estimated cell viability after 3 or 5 days' treatment with different doses of GAS (Fig. 4B) or treatment with IL-1β (10 ng/ml, 2 h) and different doses of GAS for 24 h (Fig. 4C, dark bars). One-way ANOVA showed that there were, however, significant differences in cell viability after treatment with IL-1β (10 ng/ml, 2 h) and different doses of GAS administered for 48 h ($F_{5,24} = 3.665$, $P=0.018$). *Post hoc* comparisons showed that cell viability was significantly higher in the 10 μg/ml ($P<0.01$)

and 20 μg/ml ($P<0.05$) GAS dose groups compared to the sham group.

This protective effect of GAS against disruption of NSC proliferation by IL-1β was also analyzed by BrdU incorporation. Consistent with WST-1 tests, one-way ANOVA showed that there were no significant differences between groups after GAS treatment for 24 h ($F_{5,24} = 0.096$, $P=0.983$) but there were after GAS treatment for 48 h ($F_{5,24} = 31.19$, $P=0.033$) on the ratio of BrdU⁺ to total (PI⁺) cells (Fig. 5). Finally, *post hoc* comparisons showed that BrdU⁺/PI cell ratios were significantly higher in the 10 μg/ml ($P<0.01$) and 20 μg/ml GAS dose groups compared to the sham group ($P<0.05$).

5. Discussion

Tianma has been used for thousands of years in China, but the lack safety and efficacy studies has hindered acceptance by Western medicine [34]. Phytochemical investigations have shown that GAS is one of the main phenolic compounds derived from *Gastrodia elata* Blume root [35]. Recent studies showed that GAS had some neuroprotective effects in the brain and cultured cells [36,37]. It also reported that GAS could inhibit expression of inducible NO synthase, cyclooxygenase-2 and pro-inflammatory cytokines

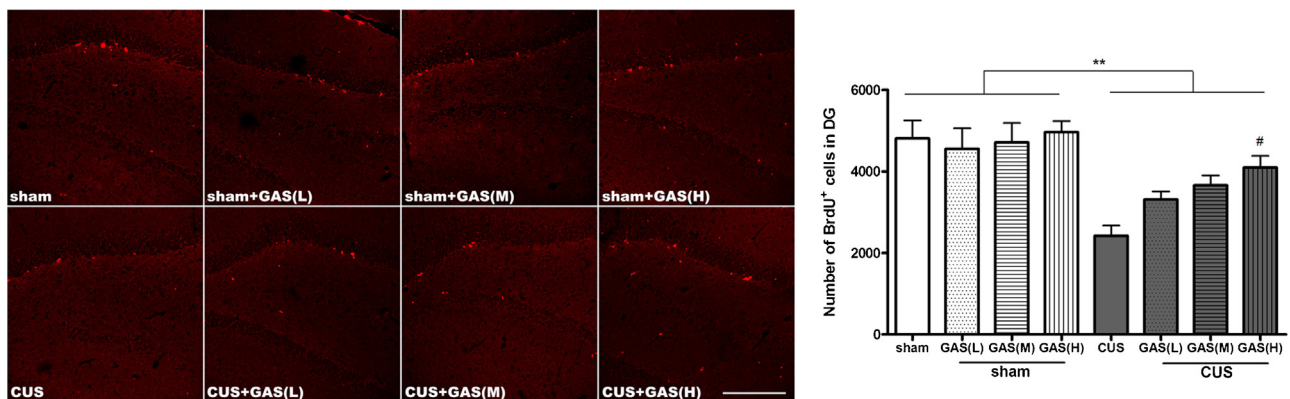


Fig. 3. GAS reversed the CUS-induced reduction in BrdU incorporation. Microphotographs (left) and histograms (right) showing BrdU-positive cells in the dentate gyrus (DG) of each group. Data were analyzed by two-way ANOVA with repeated measurement, followed by a Student–Newman–Keuls *post hoc* test. ** $P<0.05$, # $P<0.05$ vs. CUS group. Bar: 200 μm.

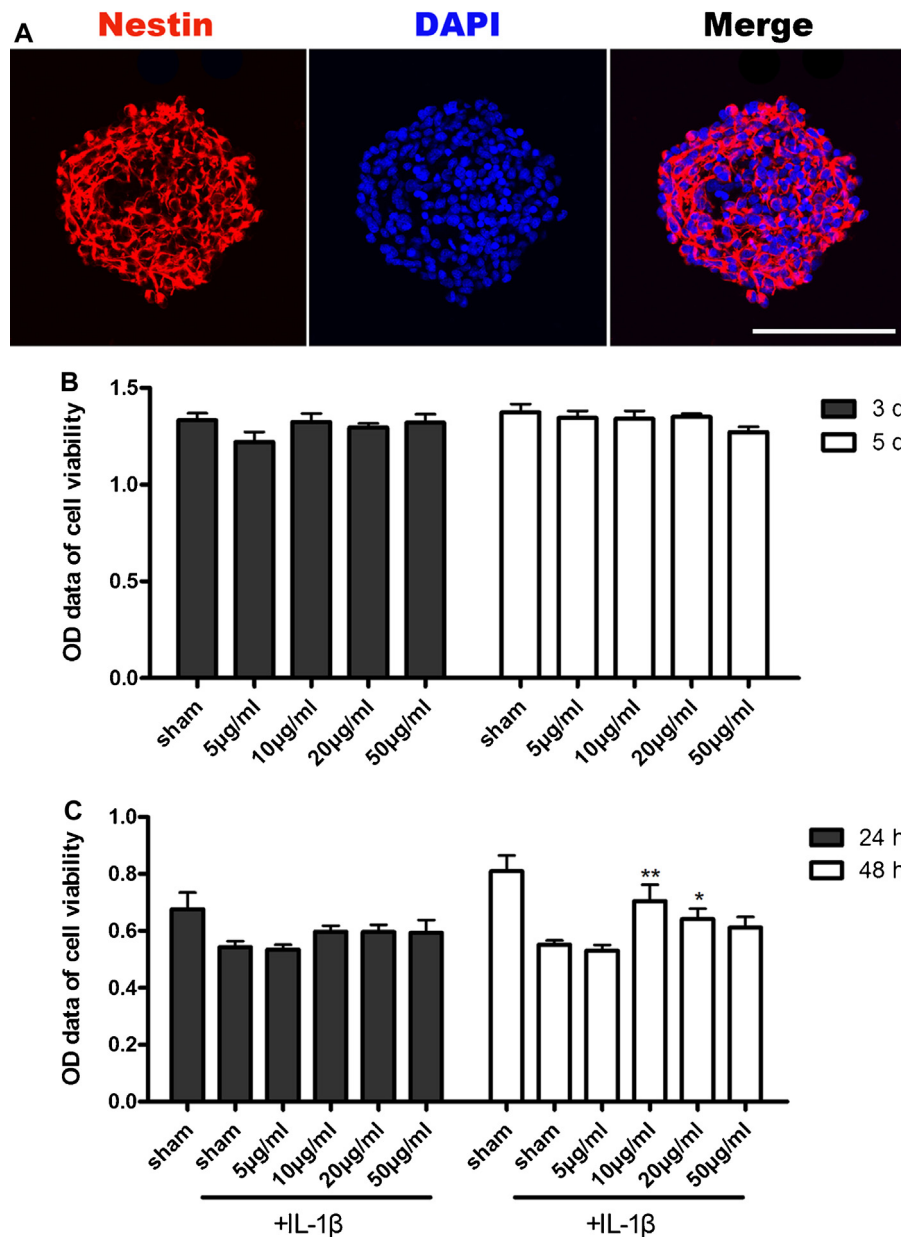


Fig. 4. Protective effect of GAS against IL-1 β toxicity in hippocampal neurosphere-derived cells *in vitro*. (A) Nestin staining of neurospheres. (B) Estimated viable cells number as measured by WST-1 assay after 3 or 5 days in different GAS concentrations. GAS had no significant effect on NSC proliferation. (C) Effect of GAS (24 or 48 h at the indicated concentrations) on viable cell number in cultures treated for 2 h with 10 ng/ml IL-1 β ($n=5$ for each group). Data were analyzed by one-way ANOVA, followed by a Least Significant Difference *post hoc* test. ** $P<0.01$ vs. sham, * $P<0.01$ vs. sham. Bar: 100 μ m.

in cultured LPS-stimulated microglia [20], and may improve learning and facilitate memory consolidation and retrieval [16]. These observations have led to the hypothesis that gastrodin may also be effective in improving psychiatric symptoms. However, little is known about its anti-depressive activity, as well as its action in the central nervous system.

In this study, we demonstrate that external stressors (CUS) induce both depression-like behaviors (anhedonia, behavioral despair, motor retardation, and spatial memory deficits) and reduced hippocampal neurogenesis in rats, consistent with our previous reports [29]. In addition, however, we show that CUS is associated with enhanced neuroinflammation as evidenced by increased IL-1 β and NF- κ B in the hippocampus. The CUS model is perhaps the most robust animal model of depression, with both construct and predictive validity. In the CUS model, animals are exposed to a series of mild and unpredictable stressors for

at least 2 weeks, which results in long-lasting behavioral, neuroanatomic, neurochemical, neuroimmune, and neuroendocrine changes [38,39]. In animal studies, many of these changes were reversed by antidepressant treatment [40]. Here we demonstrate that the natural phenol gastrodin has actions similar to clinical antidepressant in this rat model, suppressing anhedonia, learned helplessness, and cognitive disruption. Moreover, these mitigating effects may be mediated by mechanisms similar to those proposed for antidepressant in the clinic, namely preservation of hippocampus neurogenesis [41,42].

CUS decreased preference for sucrose solution in the SPT, indicating an anhedonic state, which is regarded as a core symptom of depression [43]. Daily high-dose GAS treatment (200 mg/kg) reversed the CUS-induced reduction in sucrose preference but had no effect on baseline sucrose preference in sham-treated controls, thus resembling antidepressant activity. Similarly, rats treated by

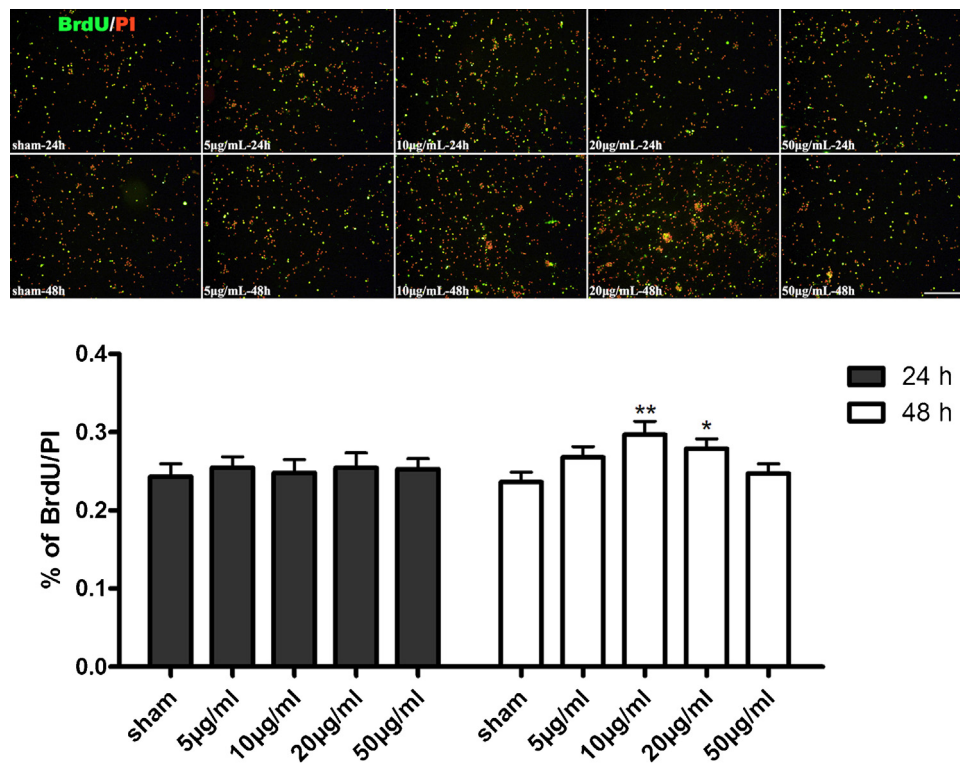


Fig. 5. Microphotographs (A) and histograms (B) showing BrdU incorporation in NSCs *in vitro* after treatment with 10 ng/ml IL-1 β and various GAS concentrations for 24 and 48 h ($n=5$ each group). Data were analyzed by one-way ANOVA, followed by a Least Significant Difference *post hoc* test. ** $P<0.01$ vs. sham, * $P<0.01$ vs. sham. Bar: 200 μ m.

CUS exhibited significant immobility in the FST, which has been interpreted as a passive stress-coping strategy (behavioral despair). This behavior was also reversed by high-dose GAS, and there is no significant difference of locomotor activity (tested by the average swimming speed of training trials in the first day of Morris water maze test) between each groups, in accord with numerous studies that have used this behavioral response to screen potential antidepressants. During and after the CUS session, stressed rats did not gain weight as quickly as sham-treated rats, but weight gain was increased to near normal by medium and high doses of GAS.

Although not direct indices of disease, clinical depression is also associated with spatial memory deficits. The MWM probe trial reflects allocentric spatial memory for platform location, and the deficits in formation of the spatial map following CUS was also ameliorated by high-dose GAS. This nootropic effect may stem from preservation of synaptoplastic processes underlying memory formation. Indeed, hippocampal long-term potentiation (LTP) is disrupted by stress and inflammation [44,45], although additional studies are required to confirm this mechanism.

Evidence is accumulating that depression is accompanied by structural changes, particularly loss of hippocampal volume, possibly due to decreased neurogenesis [46]. There is also evidence that depression is accompanied by activation of immune, inflammatory, oxidative, and nitrosative stress (IO&NS) pathways [47–49]. Hippocampal neural progenitor cells express IL-1R1 and activation of this receptor decreases cell proliferation via the NF- κ B signaling pathway [50]. Consistent with these studies, CUS impaired BrdU incorporation in the hippocampal DG, the site of hippocampal NSCs. Furthermore, expression levels of NF- κ B and IL-1 β were increased by CUS and this proinflammatory effect was reversed by GAS. To distinguish direct from indirect effects of GAS on NSC proliferation, we examined this proliferative response *in vitro*. There were no significant differences in estimated viable cell number after 3 and 5 days *in vitro* between untreated and GAS-treated cultures, so

GAS has no endogenous proliferative effects. However, GAS treatment (10 and 20 μ g/mL for 48 h) enhanced cell number in cultures treated with IL-1 β . Thus, GAS appears to prevent the impairment in NSC proliferation induced by this proinflammatory cytokine.

6. Conclusions

The results obtained from the present study suggested that GAS had therapeutic effects against depressive-like behaviors in a CUS rat model. The study also found that GAS treatment was associated with the changes in proliferation and expression of NF- κ B and IL-1 β in the hippocampus. Moreover, *in vitro* experiments revealed that GAS alone did not increase the viability of NSCs but protected them from IL-1 β -induced damage. These findings underscore the potential of GAS as a candidate natural therapy for the treatment of depression.

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